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Accuracy and reliability of artificial intelligence-assisted clinical decision-making in orthognathic surgery: A systematic review and meta-analysis

Priyawati Moungkhom, Niraj Vaghela, Hardik Ajmera, Sejal Dineshbhai Patel, Manali Prakash Jadhav

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Priyawati Moungkhom, Niraj Vaghela, Hardik Ajmera, Sejal Dineshbhai Patel, Manali Prakash Jadhav. Accuracy and reliability of artificial intelligence- assisted clinical decision-making in orthognathic surgery: A systematic review and meta-analysis. PROSPERO 2025 CRD420251143740. Available from https://www.crd.york.ac.uk/PROSPERO/view/CRD420251143740.

REVIEW TITLE AND BASIC DETAILS

Review title
Accuracy and reliability of artificial intelligence- assisted clinical decision-making in orthognathic surgery: A systematic review and meta-analysis

Condition or domain being studied
Artificial intelligence; Diagnostic Assessment; Orthognathic Assessment And Planning

Rationale for the review
Effective decision-making in orthognathic surgery necessitates rigorous discernment. It extends beyond mere clinical judgement, requiring the integration of numerous factors to be considered in the treatment planning. With the advent of advanced technologies, we can now evaluate the face in three-dimension, enabling more objective assessment. Technological advancements have made treatment planning more manageable and predictable. In particular, Artificial Intelligence (AI) has emerged as a valuable tool for assessing, planning and executing orthognathic cases. These AI based systems have superseded conventional methods in analyzing orthodontic records to anticipate the treatment outcome. However, a critical question arises, are these tools dependable? Can we truly rely on them to make a clinical decision? We therefore, proposed a systematic review to evaluate the accuracy and reliability of AI-assisted assessment in decision-making with respect to the orthognathic cases and to determine whether the predicted outcome is clinically achievable.

Review objectives

How accurate and reproducible is Artificial Intelligence (AI)-assisted assessment compared to conventional methods in making a clinical decision for orthognathic surgery?

Keywords
Artificial intelligence; Diagnostic accuracy; Orthognathic surgery; Orthodontics; Machine learning; Deep learning; AI in Dentistry; Clinical decision-making

Country
India

ELIGIBILITY CRITERIA

Population

Included

- Patients requiring or undergoing planned orthognathic treatment
- There is no restriction on age, gender or race of the patient

Excluded

- Patient with isolated orthodontic treatment without orthognathic surgery
- Cases mainly treated for trauma and reconstruction
- Syndromic craniofacial condition
- Cleft lip and cleft palate cases

Intervention(s) or exposure(s)

Included

- AI-assisted planning for orthognathic cases
- Any imaging modalities used with AI

Comparator(s) or control(s)

Included

- Conventional planning methods (clinical judgment, cephalometric analyses, diagnostic imaging)
- Studies describing only general digital planning tools without AI

Study design
Both randomized and nonrandomized study types will be included.

Included

- Descriptive, observational, cohort studies, cross-sectional, case-control studies and randomized/non-randomized trials
- Validation studies that compare AI predictions with postoperative or clinically validated outcome
- No language restriction
- Publication status- peer-reviewed journal articles
- Full-text availability

Excluded

- Case reports
- Animal studies
- Reviews, editorials or expert opinions
- Studies not involving AI
- Studies lacking measurable outcomes

Context

Effective decision-making in orthognathic surgery necessitates rigorous discernment. It extends beyond mere clinical judgement, requiring the integration of numerous factors to be considered in our treatment planning. With the advent of advanced technologies, we can now evaluate the face in three-dimension, enabling more objective assessment. Technological advancements have made treatment planning more manageable and predictable. In particular, Artificial Intelligence (AI) has emerged as a valuable tool for assessing, planning and executing orthognathic cases. These AI based systems have superseded conventional methods in analyzing orthodontic records to anticipate the treatment outcome. However, a critical question arises, are these tools dependable? Can we truly rely on them to make a clinical decision? We therefore, proposed a systematic review to evaluate the accuracy and reliability of AI-assisted assessment in decision-making with respect to the orthognathic cases and to determine whether the predicted outcome is clinically achievable.

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

25 September 2025

Review timeline

Start date: 1 January 2026. End date: 1 May 2026.

Date of registration in PROSPERO

26 September 2025

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol

A full protocol has been written and uploaded to PROSPERO. The protocol will be made available after the review is completed.

SEARCHING AND SCREENING

Search for unpublished studies

Only published studies will be sought.

Main bibliographic databases that will be searched

The main databases to be searched are *CLIB - The Cochrane Library*, *MEDLINE*, *PubMed* and *Scopus*.

Other important or specialist databases that will be searched

Google Scholar, Open-Grey

Search language restrictions
There are no language restrictions.

Search date restrictions
Databases will be searched for articles published from 1 January 2000, there are no search end date restrictions.

Other methods of identifying studies
Other studies will be identified by: *contacting authors or experts* and *looking through all the articles that cite the papers included in the review ("snowballing" or forward citation searching)*.

Link to search strategy
A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process
Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

Other relevant information about searching and screening
None

DATA COLLECTION PROCESS

Data extraction from published articles and reports
Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will be asked to provide any required data not available in published reports.

Study risk of bias or quality assessment
Risk of bias will be assessed using: *Newcastle-Ottawa*, *PROBAST* and *QUADAS-2*

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment
If ≥10 studies are available for a given outcome, funnel plots and statistical tests such as Egger's regression test will be performed where appropriate. In addition, trial registries and grey literature will be searched and assessment of certainty of evidence will be done using the GRADE approach.

Certainty assessment
Certainty of the findings will be assessed using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach. The GRADE framework considers the following domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Each outcome across the included studies will be rated as having high, moderate, low, or very

low certainty of evidence. Where quantitative synthesis is feasible, we will apply GRADE to pooled effect estimates. If meta-analysis is not possible, a narrative synthesis will be accompanied by a GRADE assessment of the certainty of evidence for each key outcome. Summary of Findings (SoF) tables will be generated to provide a transparent overview of the strength and reliability of the evidence.

OUTCOMES TO BE ANALYSED

Main outcomes

• Diagnostic Accuracy:

Sensitivity, specificity, area under the ROC curve (AUC), and predictive values of AI tools compared with conventional diagnostic/ planning methods

• Reproducibility and Reliability:

Consistency of AI- assisted decision output across datasets, observers, or repeated trials

Additional outcomes

• Clinical effectiveness and applicability:

Impact on treatment planning efficiency, error reduction, and integration into clinical workflow compared with traditional approaches.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

Narrative synthesis will be undertaken due to anticipated heterogeneity of study designs, AI models and outcome design. If ≥3 studies report with comparable methods and outcomes, a meta-analysis will be attempted.

A meta-analysis will be conducted to estimate the pooled accuracy of the model across included studies. The pooled estimate of accuracy will be calculated using a random-effects model such as DerSimonian-Laird (DL), Restricted Maximum Likelihood (REML), or Paule-Mandel (PM) that best accounts for the level of heterogeneity observed between studies.

Statistical heterogeneity will be assessed using the I^2 statistic and Cochran's Q test. If substantial heterogeneity is observed I^2 & gt; 50%, subgroup analyses or meta- regression may be performed to identify potential bias.

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage	Started	Completed
Pilot work		
Formal searching/study identification		
Screening search results against inclusion criteria		

Review stage	Started	Completed
Data extraction or receipt of IPD		
Risk of bias/quality assessment		
Data synthesis		

Review status
The review is currently planned or ongoing.

Publication of review results
Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members
Dr Priyawati Moungkhom (review guarantor and contact) ORCID: 0000-0002-6591-0588. Assistant Professor, Government dental college and hospital, Jamnagar. India.

No conflict of interest declared.

Dr Niraj Vaghela. ORCID: 0009-0005-2422-6391. Consultant Orthodontist, Ahmedabad. India.

No conflict of interest declared.

Dr Hardik Ajmera. Associate Professor, Government dental college and hospital, Jamnagar. India.

No conflict of interest declared.

Dr Sejal Dineshbhai Patel. Associate Professor, Government dental college and hospital, Jamnagar. India.

No conflict of interest declared.

Dr Manali Prakash Jadhav. Assistant Professor, Government dental college and hospital, Jamnagar. India.

No conflict of interest declared.

Named contact
Dr Priyawati Moungkhom (pandilla.pm@gmail.com). ORCID: 0000-0002-6591-0588. Assistant Professor, Government dental college and hospital, Jamnagar. India.

Review affiliation
Department of Orthodontics and Dentofacial Orthopaedics,

Government Dental College and Hospital, Jamnagar, Gujarat, India.

Funding source 1 change
Review has no specific/external funding but is supported by guarantor/review team (non-commercial) institutions.

Additional information about funding
The review is supported by the affiliated institution- Government dental college and hospital, Jamnagar

Peer review
There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Review conflict of interest
Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings
Artificial Intelligence; Clinical Decision-Making; Orthognathic Surgery; Software; Data Accuracy; Orthodontics

Revision note 1 change
to add the supporting institution for the review

SIMILAR REVIEWS

Check for similar records already in PROSPERO
PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 12 September 2025). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Applications of artificial intelligence in facial analysis for planning orthognathic surgeries: a systematic review [published 18 August 2024] [CRD42024577705]. The review was judged **not to be similar**
- Artificial Intelligence Implementation for treatment planning and Outcome Prediction in combined Orthodontic and Orthognathic Surgery – A Systematic Review of Randomized Controlled Trials [published 27 November 2022] [CRD42022366864]. The review was judged **not to be similar**
- Artificial Intelligence for Clinical Decision-Making in Neuroendocrinology: A Systematic Review [published 18 August 2025] [CRD420251129210]. **The authors did not check this review**
- "Can Artificial Intelligence algorithms enhance the performance of decision-making in dental diagnostics and treatment planning compared to conventional methods?" [published 21 February 2025] [CRD42025639340]. The review was judged **not to be similar**

PROSPERO version history 1 change

- Version 2.0, published 11 Dec 2025
- Version 1.2, published 11 Dec 2025
- Version 1.1, published 06 Dec 2025
- Version 1.0, published 26 Sep 2025

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Any enquiries about the record should be referred to the named review contact